

CARGO DRONE TRAFFIC CONTROLLER

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ASSIGNMENT TITLE : CARGO DRONE TRAFFIC CONTROLLER

Outlook of the project

This project is to control the cargo drone which controls the deliveries got from the ground stations and assign it to the drones to dispatch the request. And also maintain the overloaded drones , and the drones for rerouting.

Here I have used the array to add the delivery request's number,flight log and for dispatches to consider space constraints . Also I have used a singly linked list to add the names of the overloaded drones , a doubly linked list for the serviced drones to add the dispatch details , and can also traverse it backward to get the completed dispatches , and I have used a circular linked list to have the drones for rerouting, which can be done in a circular pattern of rerouting done

Logic of the code

Name of arrays used :

inp_arr => to add delivery request
flight_log => to add the dispatched deliveries

Name of the structures used :

Struct node => to add the overloaded drone details (singly linked list)
Struct dnode => to add the serviced drone details (doubly linked list)
Struct cnode => to add the rerouting drones details(circular linked list)

Name of the void functions used :

Void logval => to add the dispatched deliveries in a circular array
Void array => to give the output of the delivery number as the kind of delivery
Void enqueue => to add the deliveries to the queue
Void dequeue => to dispatch deliveries normally from the front
Void pop => to make priority dispatch , removing the recent element added
Void createNodelist => to add the overloaded drone names to the structure
Void insert => to add the serviced drone names to the structure
Void addnode => to add rerouting drone names to the structure
Void deletenode => to remove the name of the rerouted drone

The main function starts with a while loop and a switch case which gives 5 options to choose like the delivery request, dispatch, flight log display, maintenance tracker or exit option.

When the user gets into the delivery request option there is also a variety of options to choose the kind of delivery required

The second option has the access to either normal dispatch or the priority dispatch and assigning it to one of the drones in the service drone list

The third option only displays the recent 6 flight log , the requests dispatched

The fourth option has options to choose between the add or display names of the overloaded drones, add or display names of the serviced drones , add or display the names of drones for rerouting , release drone when condition is right , and last with the display of recent dispatches done along with drone names and the request dispatched .

Design explanation

This design is mainly to reduce time complexity and space reduction , and with simple codes for easy understanding . I have used the arrays to stores only the integer values , but while printing gives the output as the delivery which I made with another void function.

I have also didn't create any new list or structure to store the dispatched items instead directly printing it with the help of traversing the doubly linked list , mainly using the advantage of the doubly linked list .

Output screenshots

```
Output Clear  
enter the service to be provided  
1.add delivery request  
2.dispatch  
3.flight log unit  
4.maintenance tracker  
5.exit  
|
```

```
enter the service to be provided  
1.add delivery request  
2.dispatch  
3.flight log unit  
4.maintenance tracker  
5.exit  
1  
enter the delivery request required1. food  
2.medicine  
3.tools  
4.water  
5.parts  
6.fuel  
1  
request accepted
```

enter the service to be provided

1.add delivery request

2.dispatch

3.flight log unit

4.maintenance tracker

5.exit

2

is it a priority dispatch or normal way dispatch??if priority press 1 or press any other number:1

request in progress is tools

assigning the dispatch to drone named : drone_a

enter the service to be provided

1.add delivery request

2.dispatch

3.flight log unit

4.maintenance tracker

5.exit

4

maintenance tracker

enter a choice

1.overloaded drones

2.seviced drones

3.emergency drones

4.get list of recently dispatched drones2

do u want to 1.display names or 2. input names

2

enter the name of the drone to include

drone_a

```
enter the service to be provided
1.add delivery request
2.dispatch
3.flight log unit
4.maintanence tracker
5.exit
4
maintanence tracker
enter a choice
1.overloaded drones
2.seviced drones
3.emergency drones
4.get list of recently dispatched drones3
select one option
1.add drone
2.release drone
3.display drones1
enter the name of emergency drone to be rerouted:
drone_b
```

```
enter the service to be provided
1.add delivery request
2.dispatch
3.flight log unit
4.maintanence tracker
5.exit
3
Displaying the last 1 dispatches done
tools
```

```
enter the service to be provided
1.add delivery request
2.dispatch
3.flight log unit
4.maintanence tracker
5.exit
4
maintanence tracker
enter a choice
1.overloaded drones
2.seviced drones
3.emergency drones
4.get list of recently dispatched drones3
select one option
1.add drone
2.release drone
3.display drones2
the emergency drone that is rerouted is : drone_a
```

```
1.add delivery request
2.dispatch
3.flight log unit
4.maintanence tracker
5.exit
4
maintanence tracker
enter a choice
1.overloaded drones
2.seviced drones
3.emergency drones
4.get list of recently dispatched drones3
select one option
1.add drone
2.release drone
3.display drones3
drone_b
```